IEEE VIS 2026 Tutorial

Introduction to Quantum Computing for Visualization Researchers

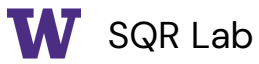
Organizers

Hyeok Kim
KAIST

Sara Mouradian
University of Washington

Leilani Battle
University of Washington







Session 2 (Part 3/4)

Part 3

Recent HCI/VIS research in Quantum Computing

Topics covered

Understanding quantum states

Programming support

Working with quantum machines

Interpreting outcomes

Domain-specific support

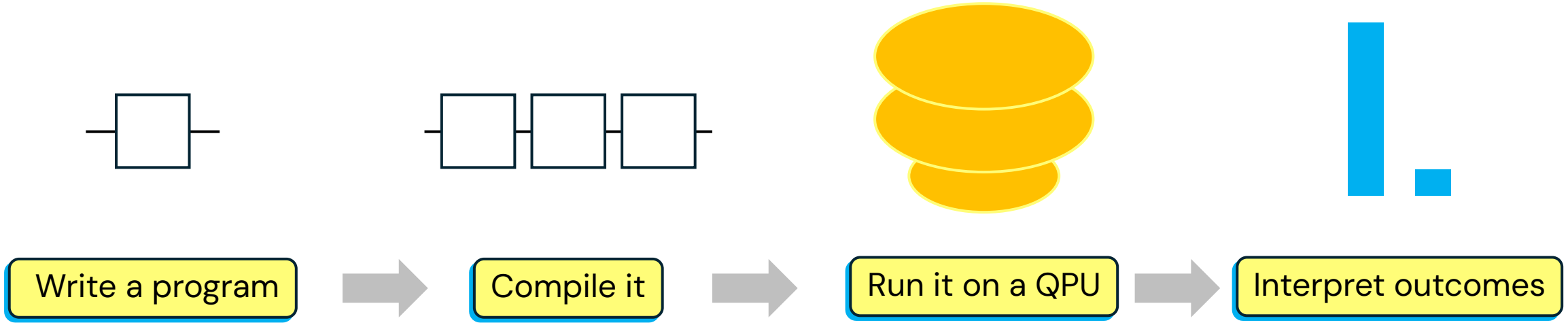
Education research

User-side workflow for quantum computing

Simplified

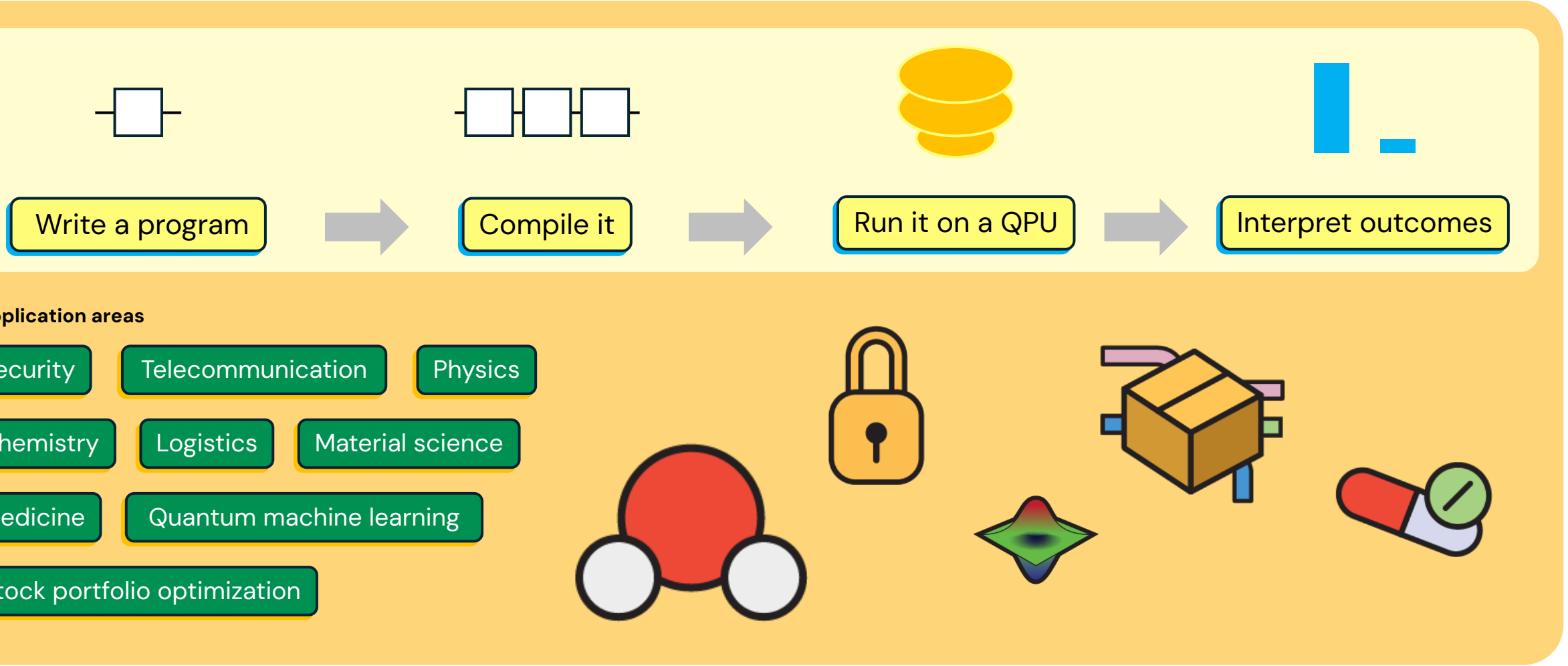
User-side workflow for quantum computing

Simplified

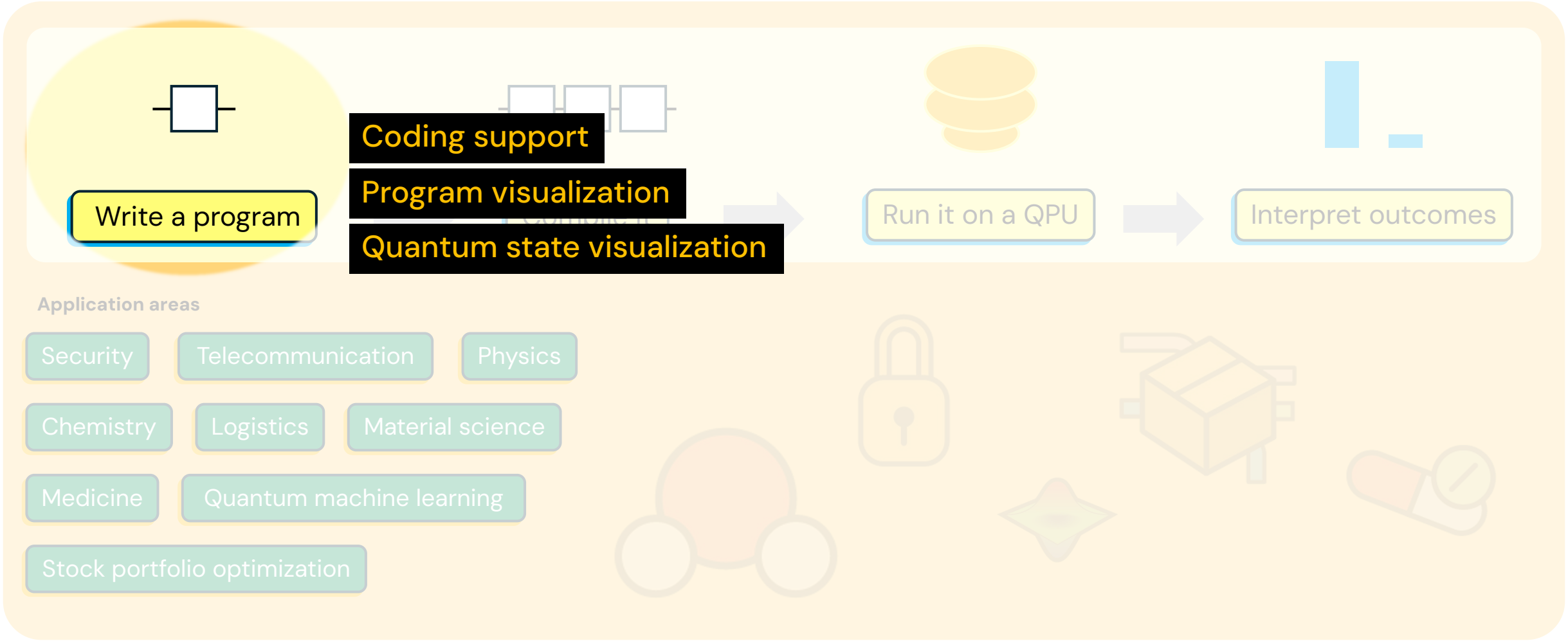


User-side workflow for quantum computing

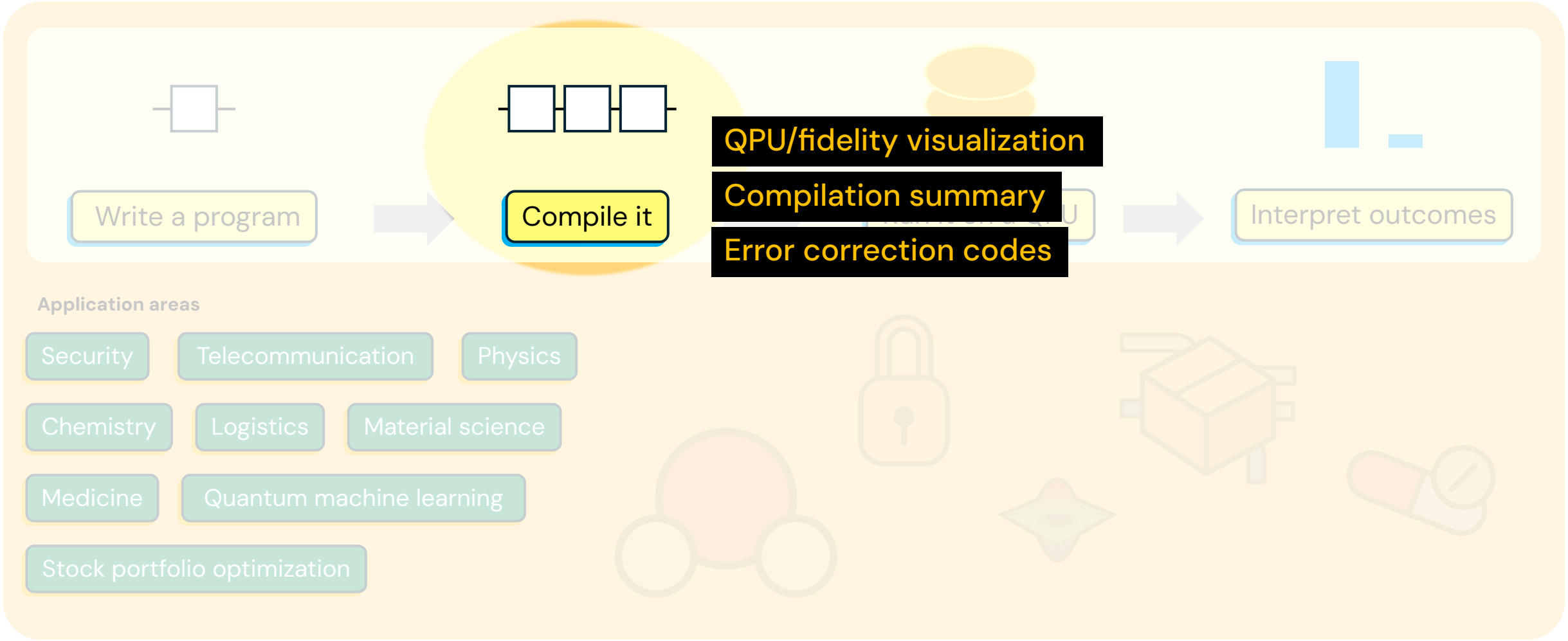
Simplified



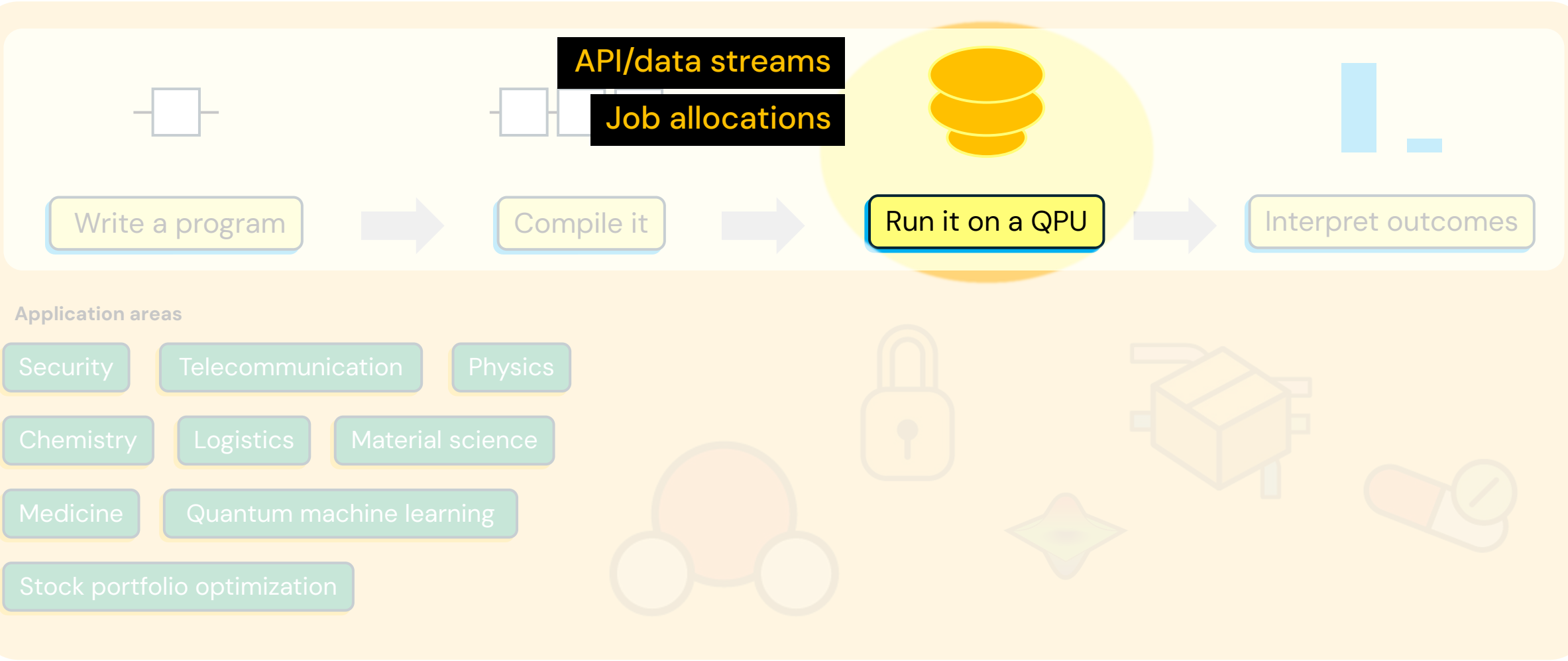
HCI/VIS interests



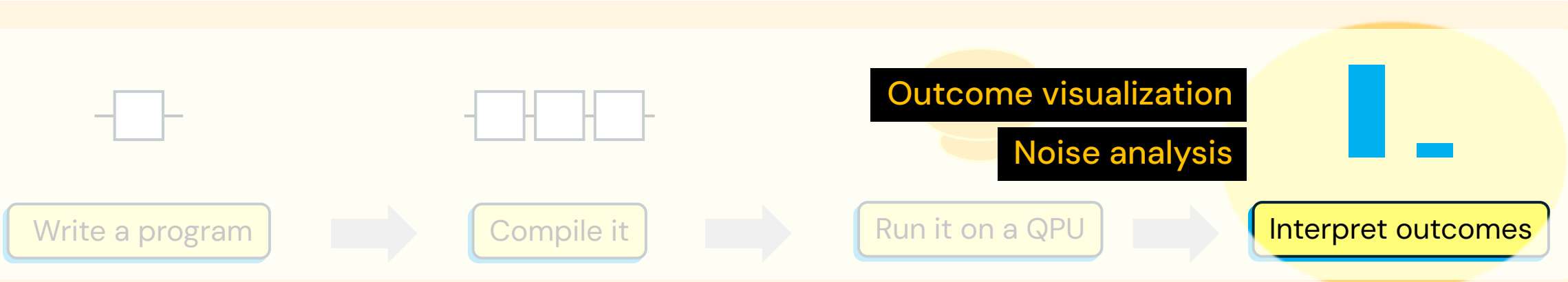
HCI/VIS interests



HCI/VIS interests



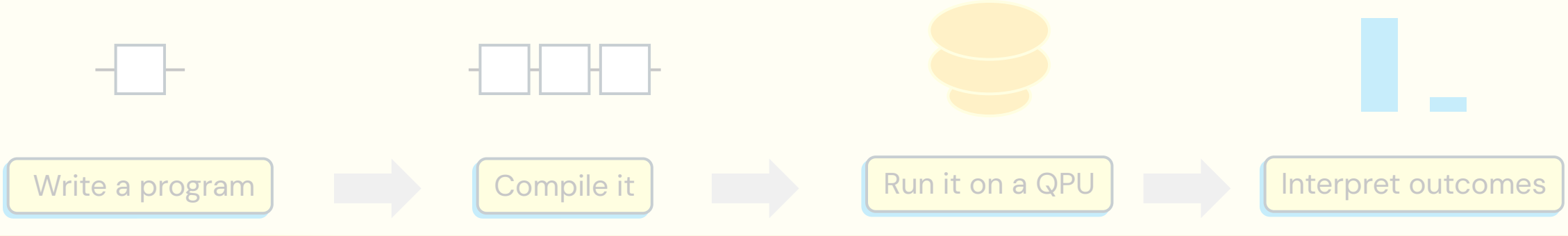
HCI/VIS interests



Application areas

- Security
- Telecommunication
- Physics
- Chemistry
- Logistics
- Material science
- Medicine
- Quantum machine learning
- Stock portfolio optimization

HCI/VIS interests



Application areas

- Security
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Domain-specific analytic interfaces

Mapping algorithms to real-world problems

Program support

Coding support

Quantum state visualization

Program visualization

Program support

Coding support

Quantum state visualization

Program visualization

Quirk: Drag and drop

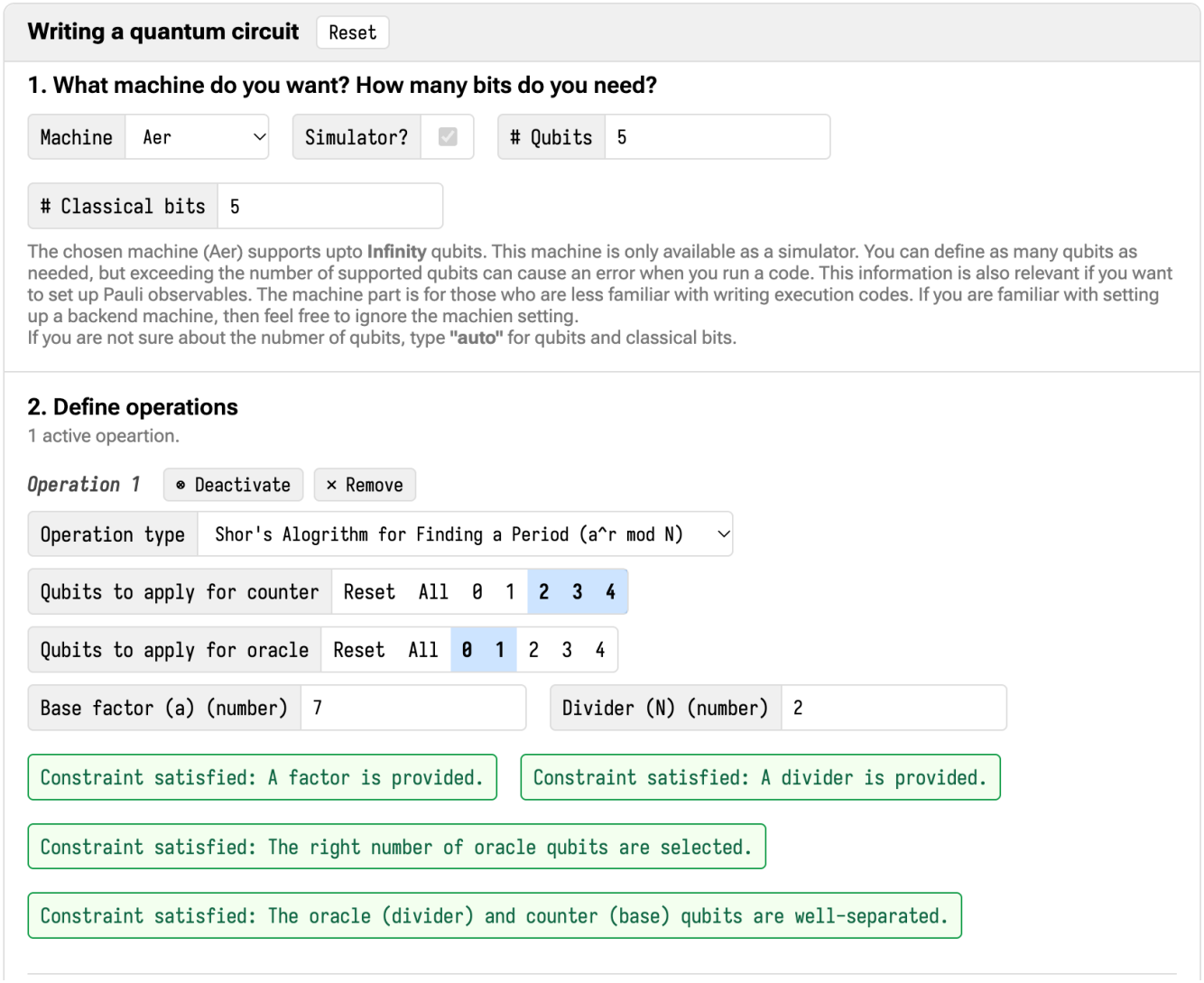
[Gidney et al., 2019]

https://algassert.com/quirk

Scalability limits

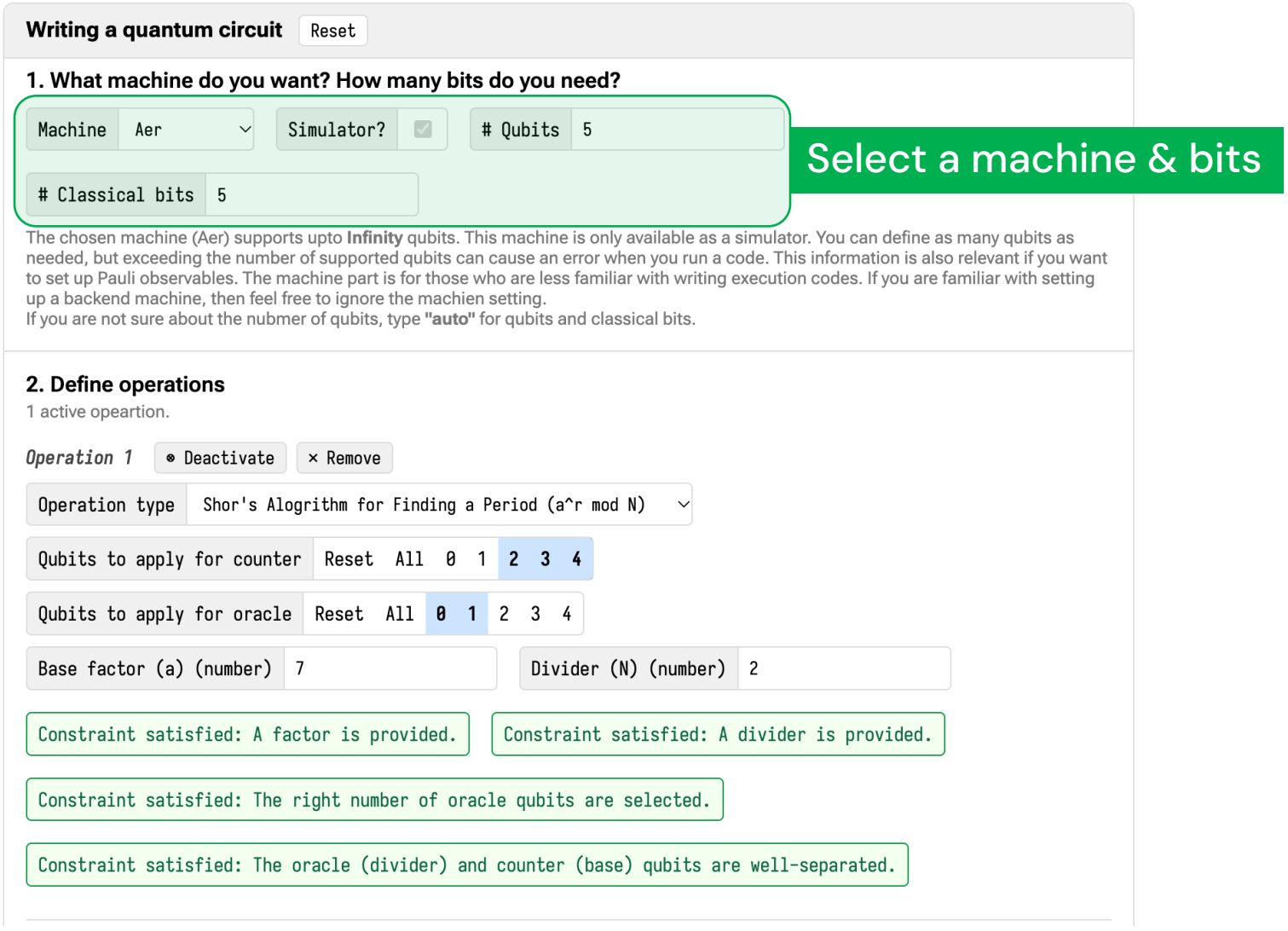
Coding support

[Kim et al., 2025]



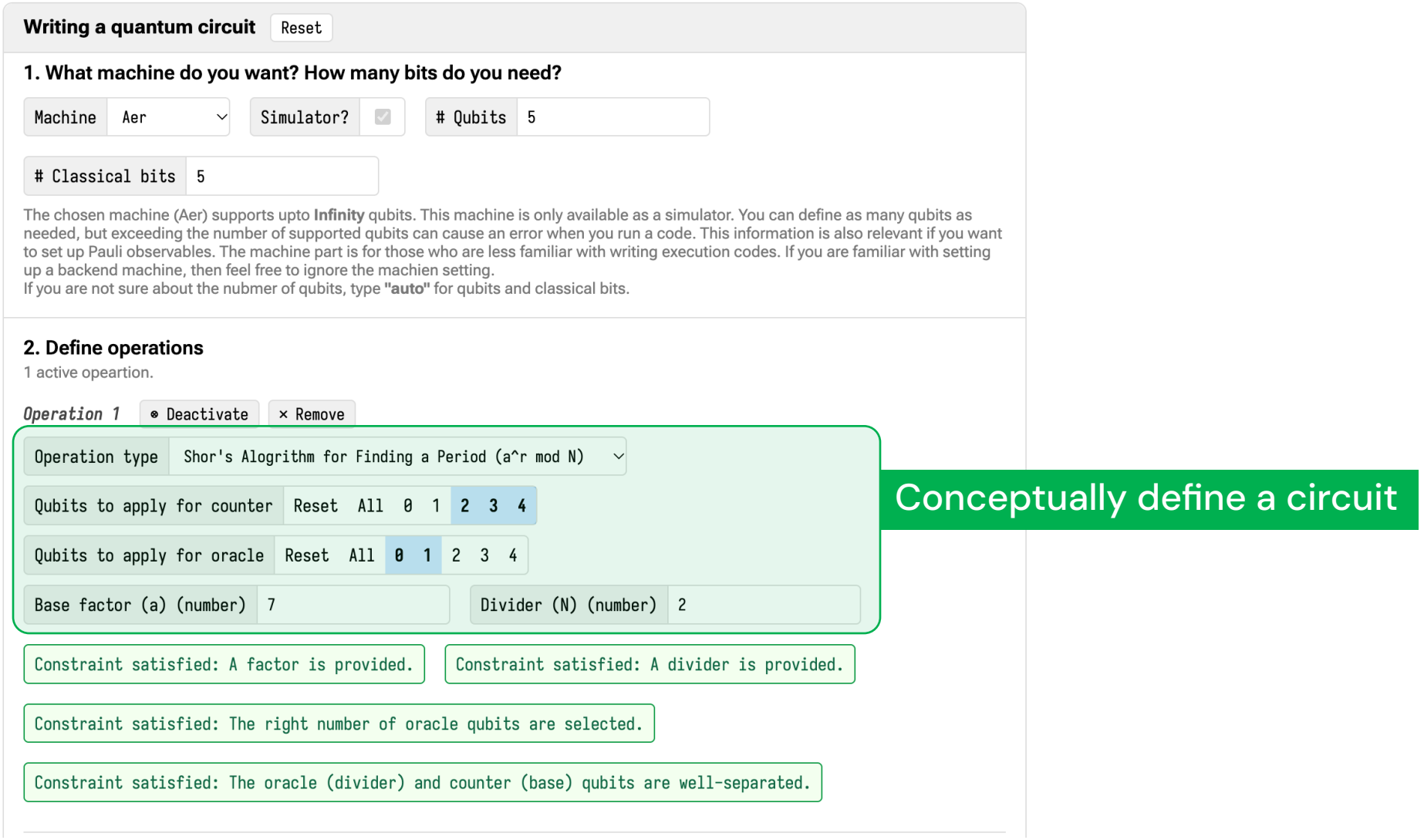
Coding support

[Kim et al., 2025]



Coding support

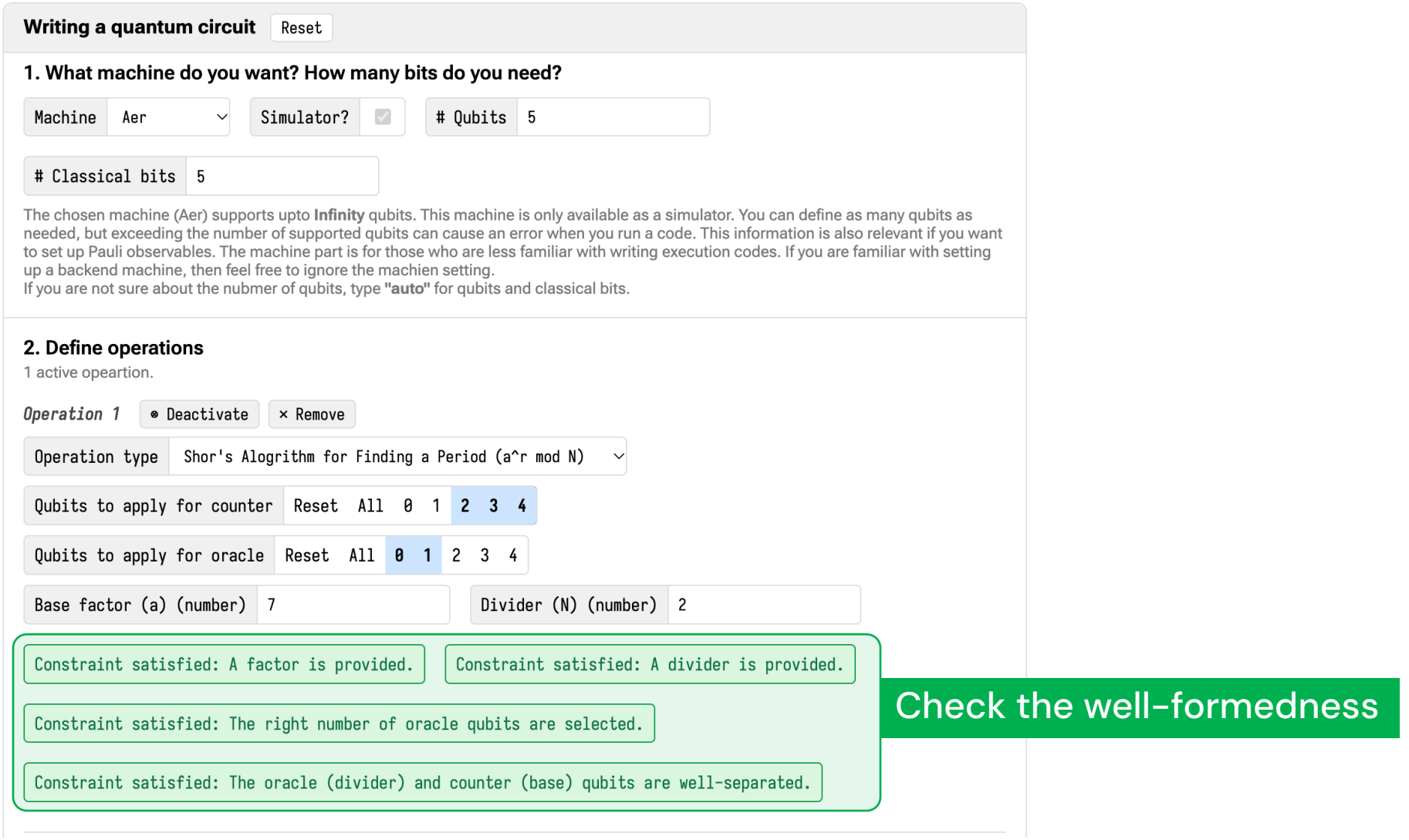
[Kim et al., 2025]



Conceptually define a circuit

Coding support

[Kim et al., 2025]



Check the well-formedness

Coding support with templates

[Classiq]

You can use the free demo after creating an account
https://www.classiq.io/

APIs for domain-specific algorithms

[Microsoft]

```python
import numpy as np
from qdk_chemistry.data import Structure

# Specify a structure
coords = np.array([[0.0, 0.0, 0.0], [0.0, 0.0, 1.4]])
symbols = ["H", "H"]
structure = Structure(coords, symbols=symbols)

# Run scf
scf_solver = create("scf_solver")
E_scf, wfn_scf = scf_solver.run(
    structure, charge=0, spin_multiplicity=1, basis_or_guess="sto-3g"
)

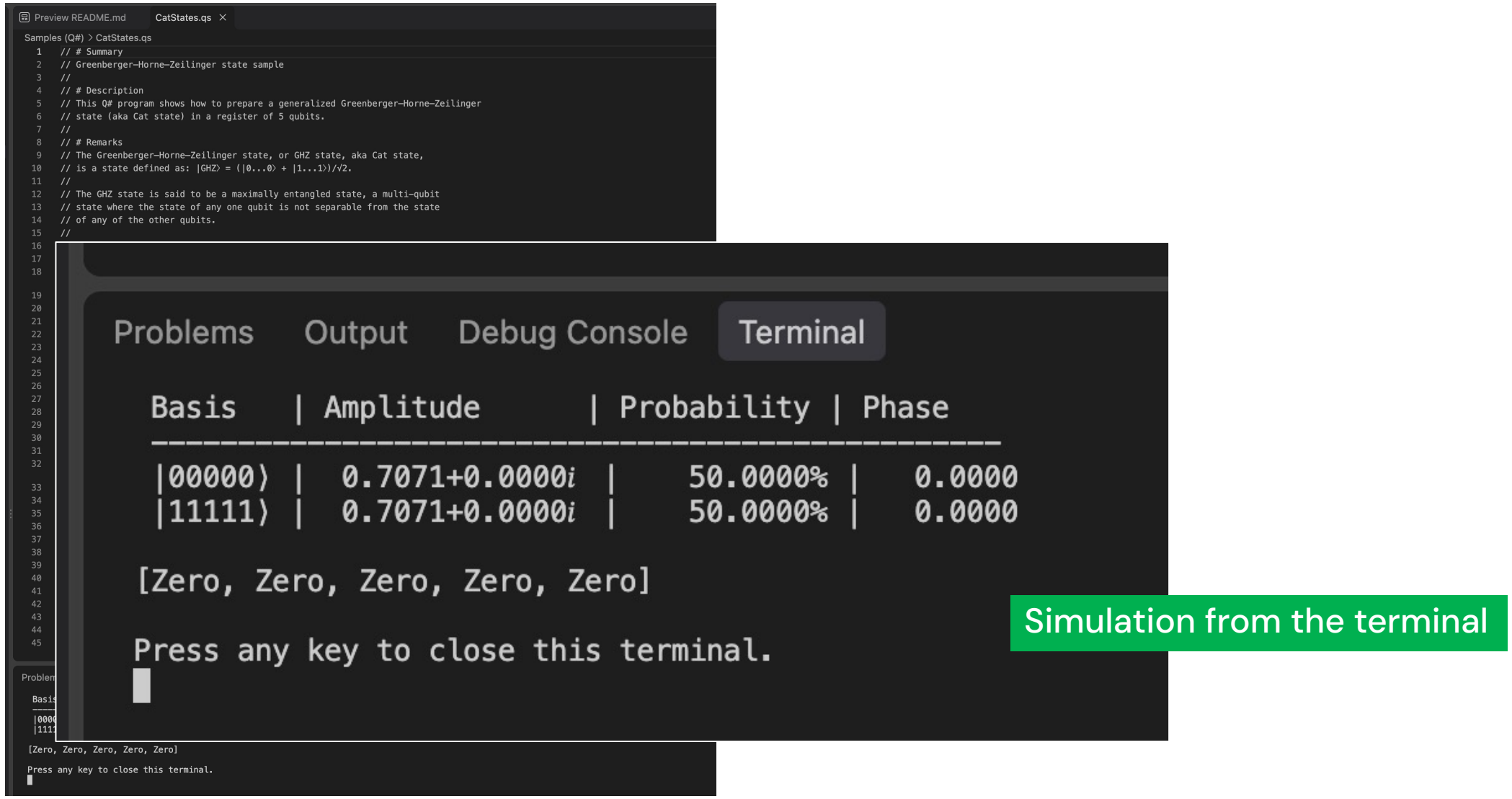
# Compute the Hamiltonian
hamiltonian_constructor = create("hamiltonian_constructor")
hamiltonian = hamiltonian_constructor.run(wfn_scf.get_orbitals())

# Compute CAS wavefunction
cas_solver = create("multi_configuration_calculator", "macis_cas")
E_cas, wfn_cas = cas_solver.run(hamiltonian, 1, 1)

# Construct the circuit
regular_circuit = regular_prep.run(wfn_cas)
sparse_circuit = sparse_prep.run(wfn_cas)
dense_circuit = dense_prep.run(wfn_cas)
print(f"Regular isometry circuit:\n{regular_circuit.get_qiskit_circuit()}")
print(f"Sparse isometry circuit:\n{sparse_circuit.get_qsharp_circuit()}")
print(f"Dense pure state circuit:\n{dense_circuit.get_qsharp_circuit()}")
```

In-IDE support

[Microsoft]



Simulation from the terminal

QuantumKatas | IDE as a textbook

[Microsoft, Archived]

https://github.com/microsoft/quantumkatas

LLM assistants

[Dupuis et al., 2024; IBM]

https://ieeexplore.ieee.org/abstract/document/10691762/

Microsoft has something
similar with GitHub Copilot.

Code verification for LLMs

[Dupuis et al., 2025; IBM]

https://arxiv.org/abs/2508.20907

Compute the ratio of the successful unit tests per a code generated
How we synthesize unit tests is critical

Qiskit-HumanEval

[IBM]

https://github.com/qiskit-community/qiskit-human-eval

Education support redirected to LLM assistants

[IBM]

https://arxiv.org/abs/2605.27210

Program support

Coding support

Quantum state visualization

Program visualization

BEADS | Colored Bloch sphere

[Huber & Glaser, 2024]

Looking a Bloch sphere
through the lens of measurement device

Similar to diverging scales



Likely to be
0

0 or 1

1

Figure 2
https://arxiv.org/abs/2410.01446

BEADS | Colored Bloch sphere

[Huber & Glaser, 2024]

Expressing the effects
of different gates

Figure 3
https://arxiv.org/abs/2410.01446

BEADS | Colored Bloch sphere

[Huber & Glaser, 2024]

Expressing entanglement probabilities

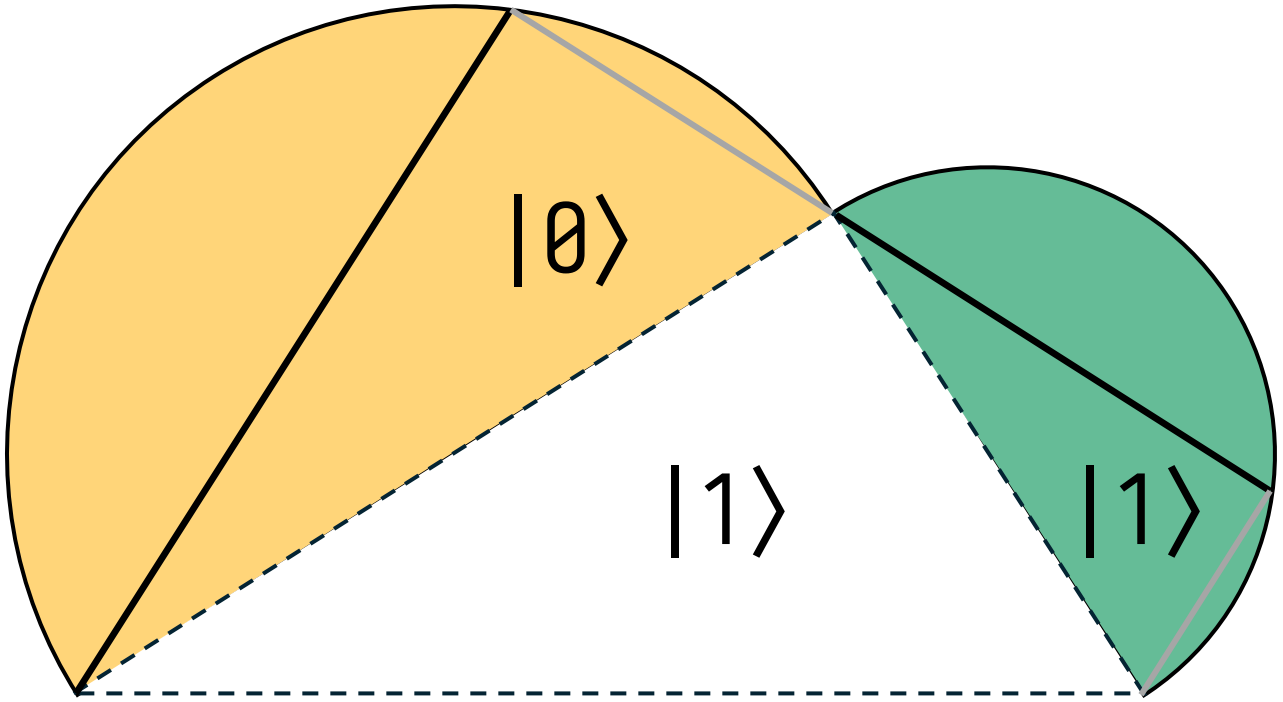
Venus | Geometric representation

[Ruan et al., 2023]

https://onlinelibrary.wiley.com/doi/abs/10.1111/cgf.14827

Venus | Geometric representation

[Ruan et al., 2023]



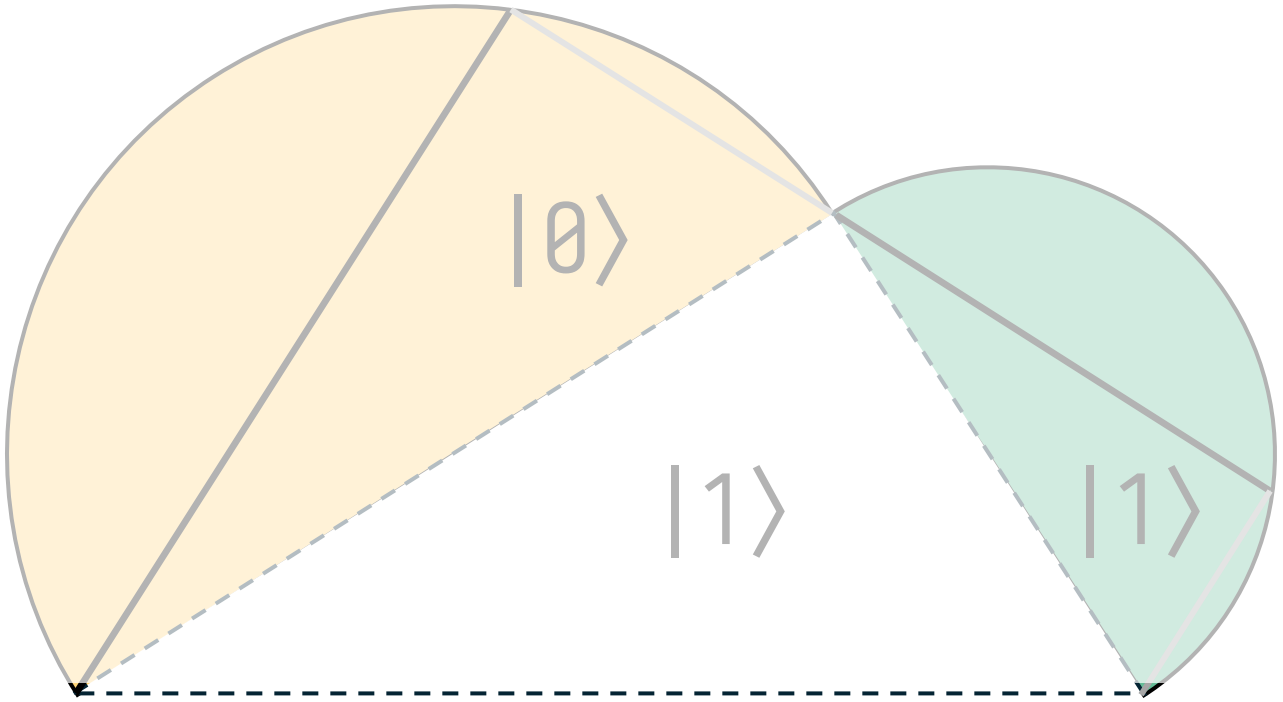
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\alpha = a_1 + a_2 i$$

$$\beta = b_1 + b_2 i$$

Venus | Geometric representation

[Ruan et al., 2023]



= **1** (normalization constraint)

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$|\alpha|^2 + |\beta|^2 = 1$$

$$\alpha = a_1 + a_2 i$$

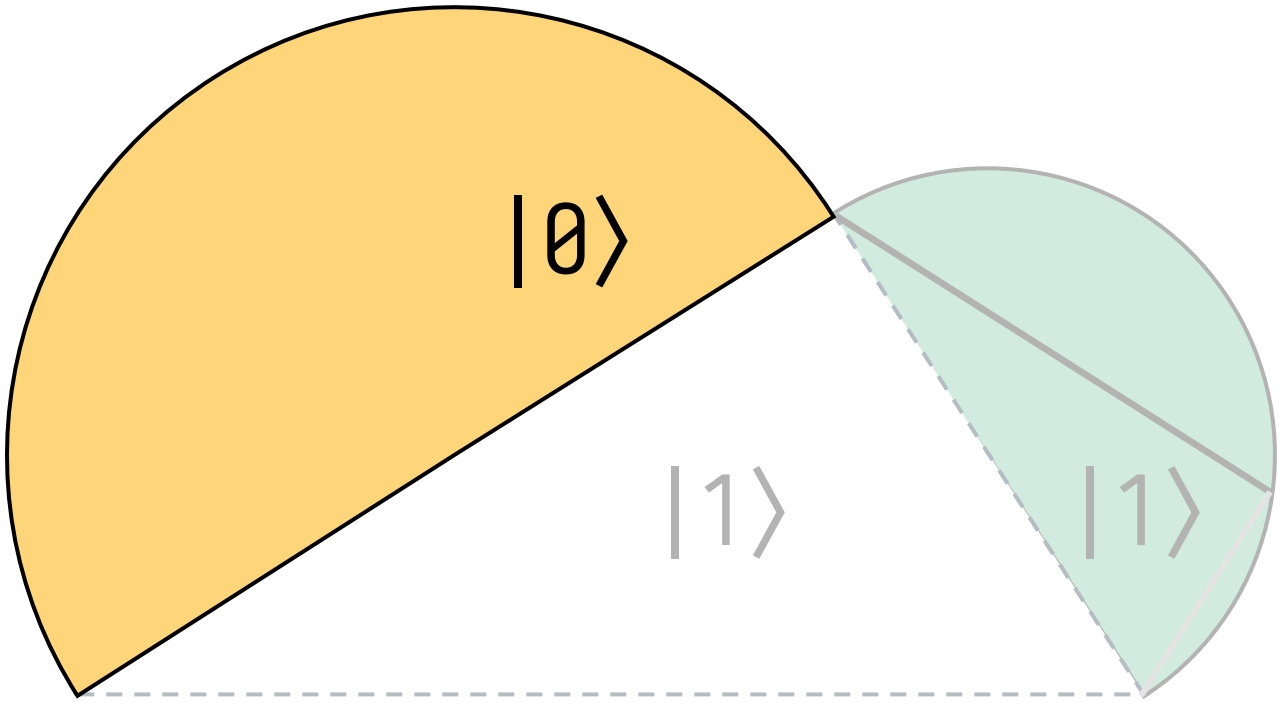
$$\beta = b_1 + b_2 i$$

← Pythagorean theorem!

Venus | Geometric representation

[Ruan et al., 2023]

= probability of measuring 0



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\alpha = a_1 + a_2 i$$

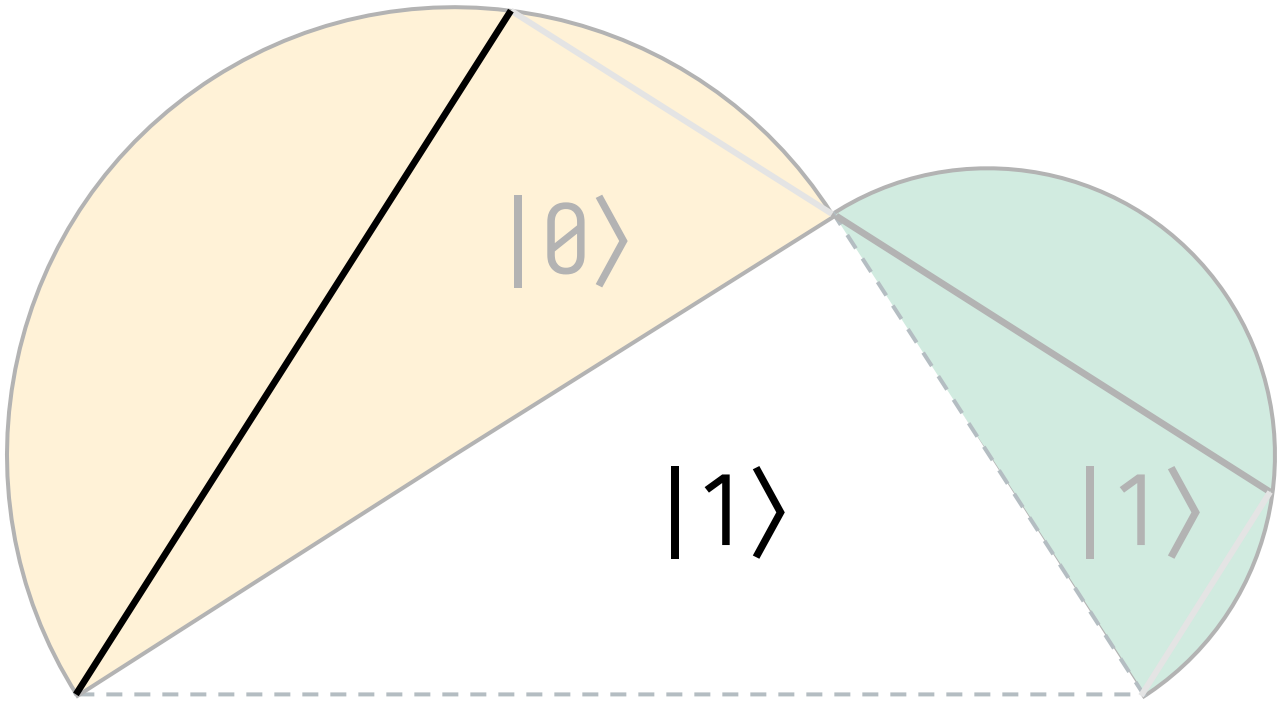
$$|\alpha|^2$$

$$\beta = b_1 + b_2 i$$

Venus | Geometric representation

[Ruan et al., 2023]

real part of amplitude =



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\alpha = a_1 + a_2 i$$

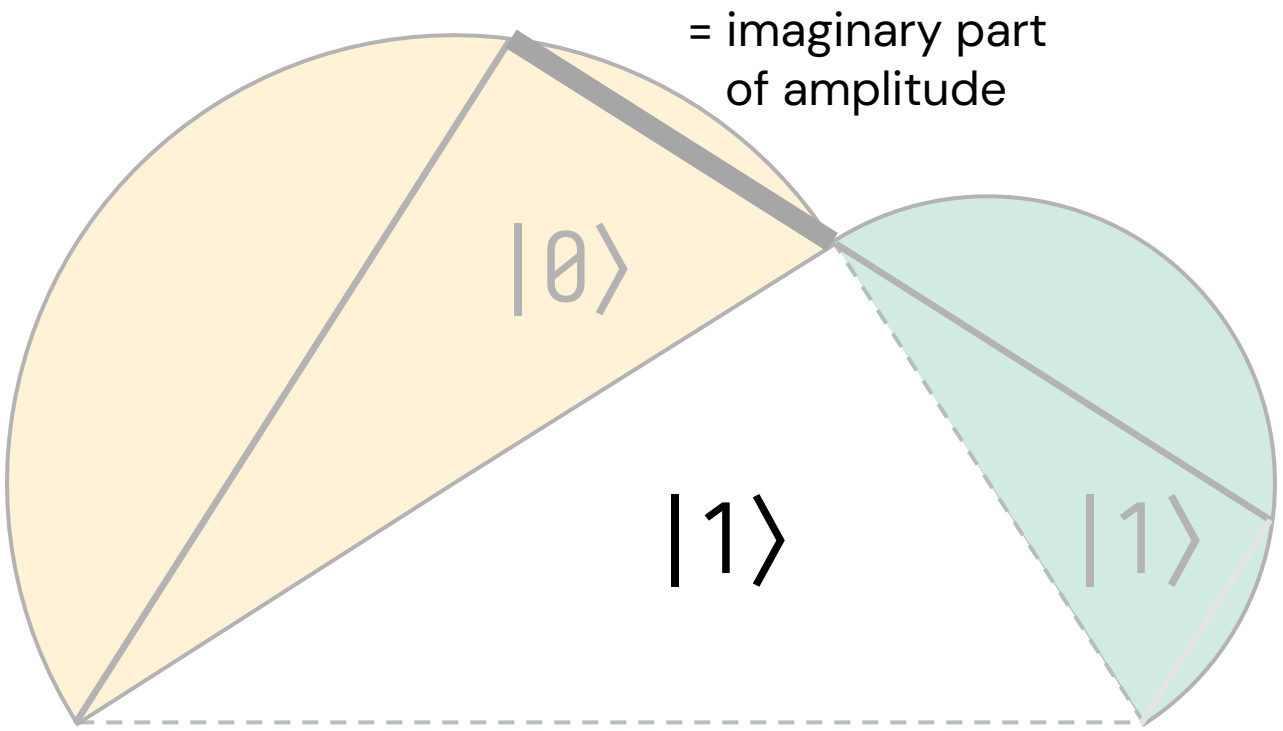
$$|\alpha|^2 = (a_1 + a_2 i)(a_1 + a_2 i)$$

$$= a_1^2 + a_2^2$$

$$\beta = b_1 + b_2 i$$

Venus | Geometric representation

[Ruan et al., 2023]



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\alpha = a_1 + a_2 i$$

$$|\alpha|^2 = (a_1 + a_2 i)(a_1 + a_2 i)$$

$$= a_1^2 + a_2^2$$

$$\beta = b_1 + b_2 i$$

Program support

Coding support

Quantum state visualization

Program visualization

Changes to multi-qubit states

[Ruan et al., 2023]

https://ieeexplore.ieee.org/abstract/document/10319321

How probabilities for multi-qubit states (00, 01, ...) change over operations

Quantivine | Toward scalability

[Wen et al., 2024]

https://ieeexplore.ieee.org/abstract/document/10296051

Grouping operations by their functions

Inspecting operations on a single qubit

Declutter repetitive information

Operation distribution on a circuit

QuFlow | Gate parameter changes

[Lin et al., 2018]

https://ieeexplore.ieee.org/abstract/document/8823602

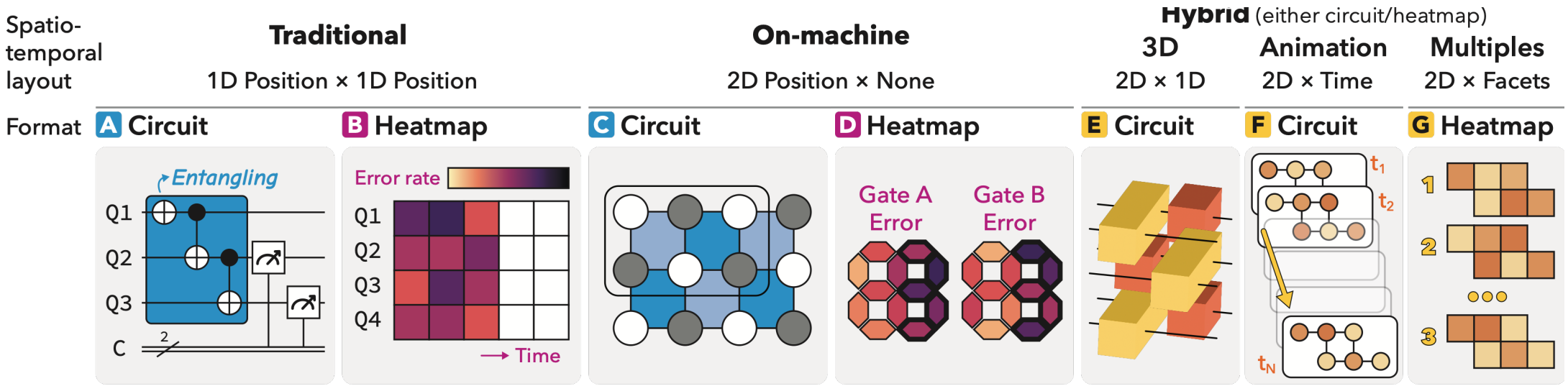
Statevector changes over a program

[McGuffin & Robert, 2026]

https://dl.acm.org/doi/abs/10.1145/3786463

Circuit visualization design space

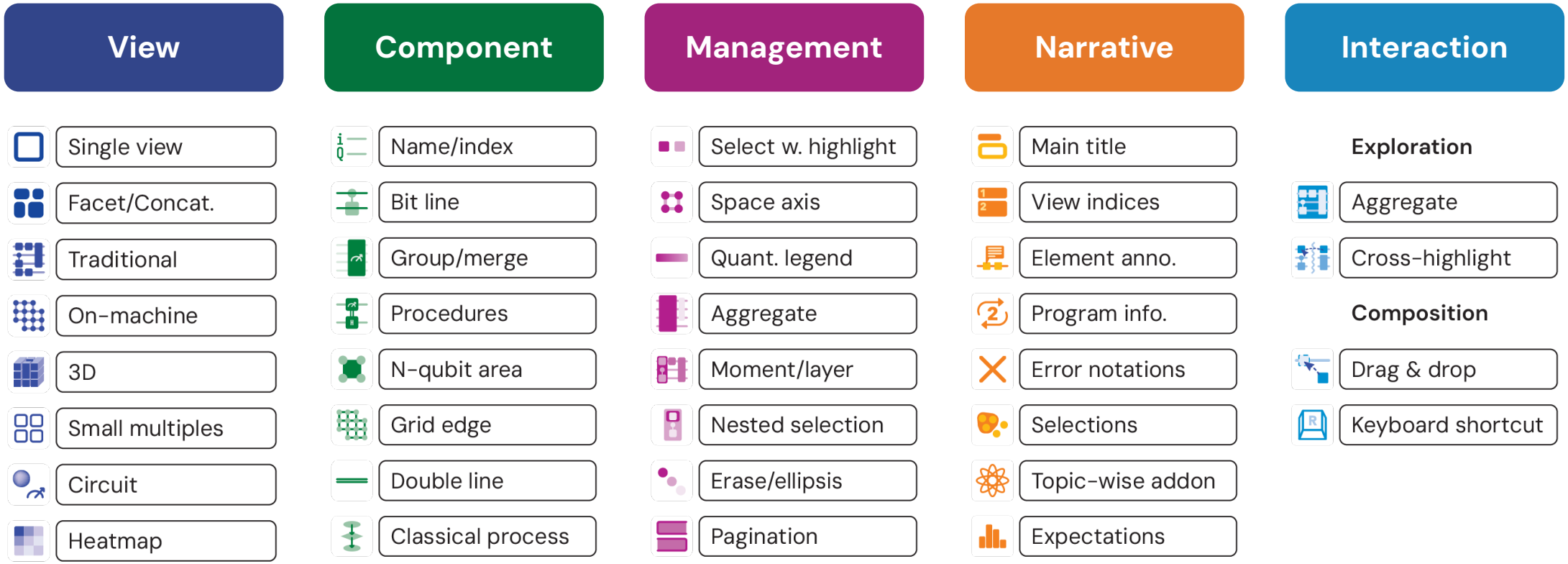
[Kim et al., 2026]



Presentation → Thursday 1pm @ Hall Essex north

Circuit visualization design space

[Kim et al., 2026]



Compiling quantum circuits

QPU/fidelity visualization

Compilation summary

Error correction codes

Compiling quantum circuits

QPU/fidelity visualization

Compilation summary

Error correction codes

Fidelity

[IBM Quantum Platform]

Qubit:

Readout assignment error

Median 9.583E-3

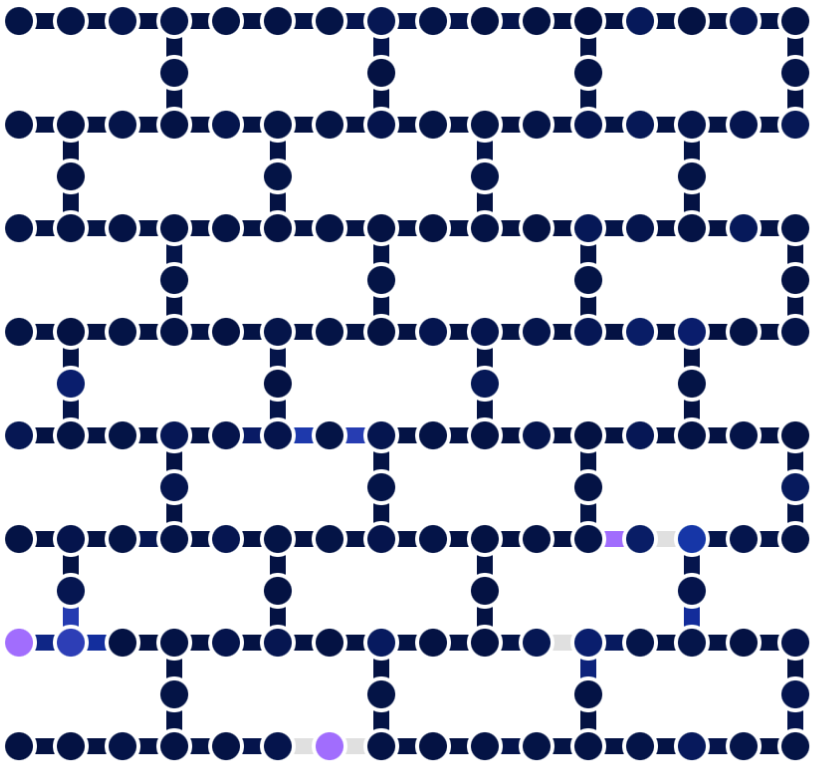
min 2.686E-3 max 4.907E-1

Connection:

CZ error

Median 2.125E-3

min 1.133E-3 max 1.639E-1



VACSEN | Noise awareness

[Ruan et al., 2023]

https://ieeexplore.ieee.org/abstract/document/9904430

Error rate distribution per machine

Compilation distribution

Compilation details

Compiling quantum circuits

QPU/fidelity visualization

Compilation summary

Error correction codes

VACSEN | Noise awareness

[Ruan et al., 2023]

https://ieeexplore.ieee.org/abstract/document/9904430

Compare summary fidelity

Estimated success probability

[Nishio et al., 2020]

$$SR_i = 1 - error_i,$$

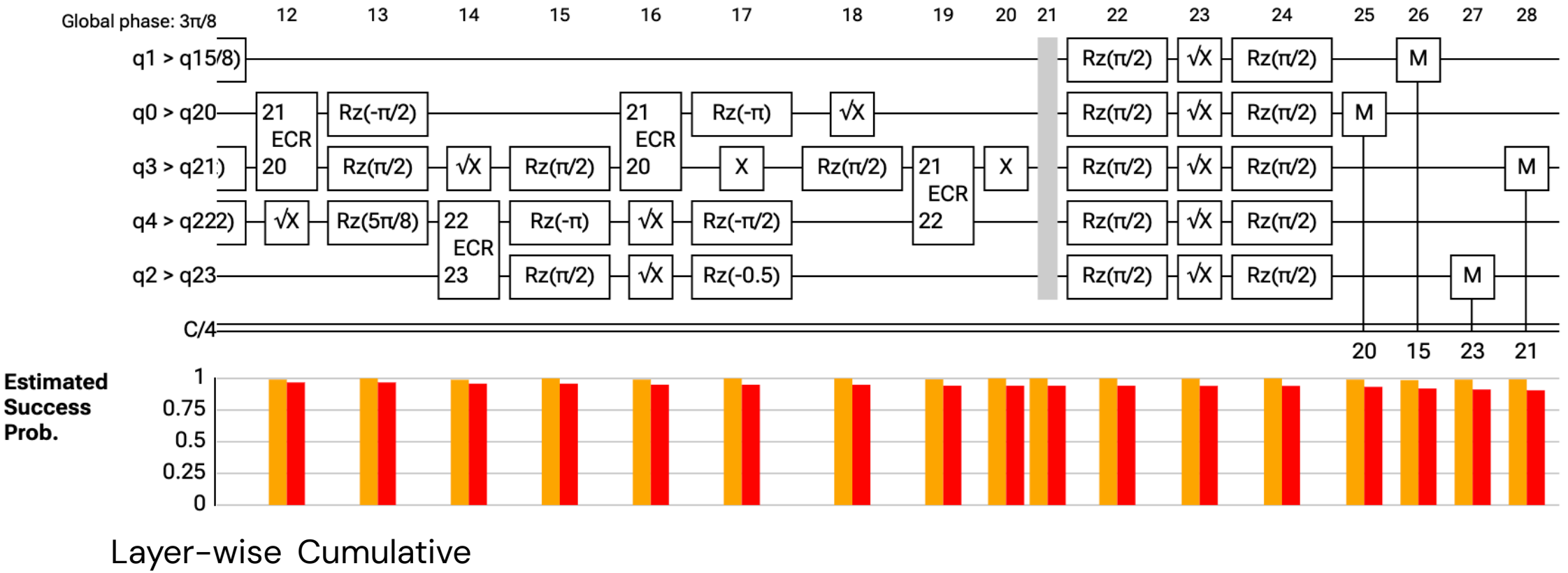
$$ESP = \prod_i SR_i\,,$$

$$i\text{: all the gates in a given circuit}$$

SR: success rate

Estimated success probability

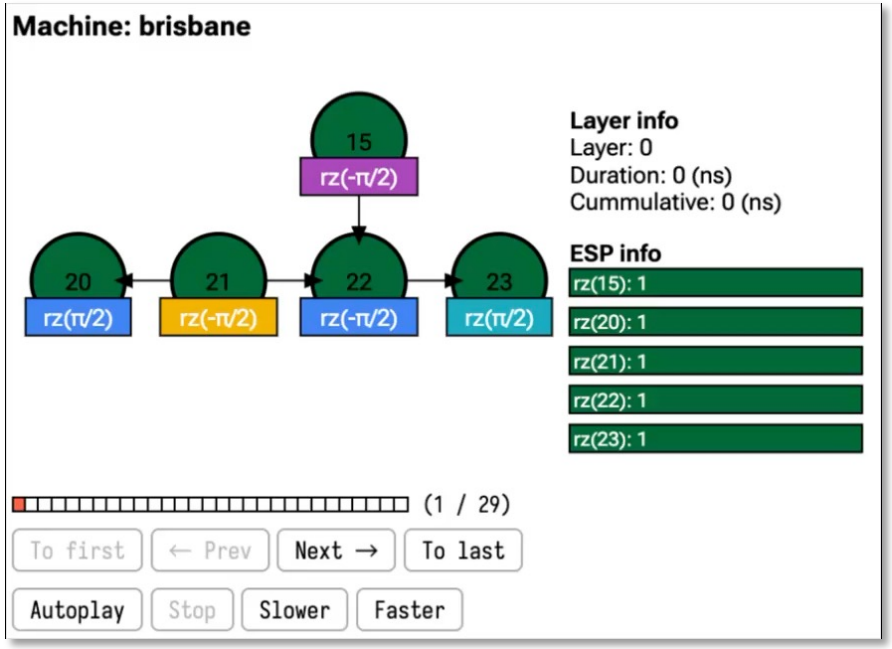
[Kim et al., 2025]



Layer-wise Cumulative

Compilation results on a QPU

[Kim et al., 2025]



Compiling quantum circuits

QPU/fidelity visualization

Compilation summary

Error correction codes

Crumble | Error correction support

[Gidney, 2021]

Showing one layer at a time

How implemented on a QPU

Entire view

Operations for QEC

https://algassert.com/crumble

Crumpy | Error propagation

[Zappa et al., 2025]

https://github.com/guenp/crumpy

The red triangle denotes where a Pauli X (bitflip) error has happened.
The red lines denote how that error propagate through the circuit.

Time vs. Space

Quantum circuits have three primary dimensions at a physical level.

Dimensions	2D space on a QPU	1D time progression
Why matter?	Connectivity information	How circuits evolve over time

Working with QPUs

API/data streams

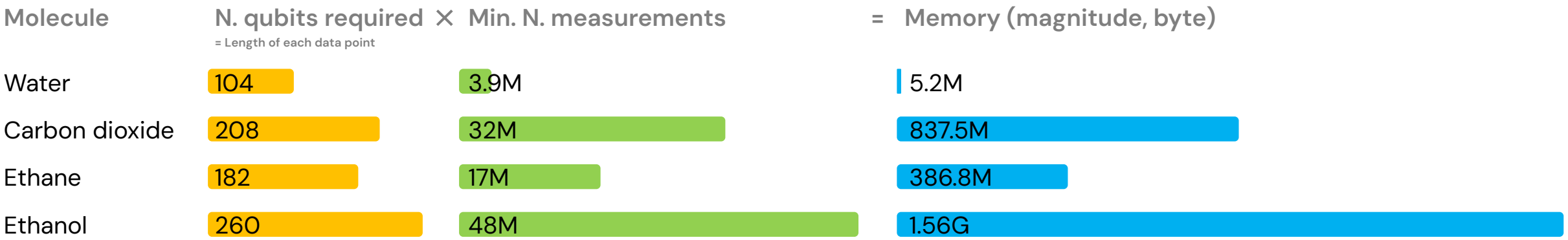
Job allocations

Sending a circuit to a machine itself is pretty straightforward and easy.
However, how we use those resources is quite tricky and needs attention.

Streaming large data

At least right now, we cannot observe outcomes of quantum computers in-situ,
but we may need to load them to a classical computer at a slower pace.

For some algorithms, such as Shor's and Grover's, would not cause such problems,
but others like chemical problems like VQE would (Gonthier et al., 2022).



To scale, an ethanol molecule has 9 atoms while the number is 33 and 788 for ibuprofen and insulin, respectively.

Allocating jobs

Many people are using the same quantum computers.

Machine	Region	Programmable qubits	Physical qubits	Status	N. Pending jobs
ibm_pittsburgh	us-east	156	332	Online Maintenance: 10/26 - 10/27	1,039

Your job needs 3 seconds, but you might need to wait a day or two.

IBM's fair-share scheduling

Among the pending jobs, scheduler prefers

- "least requested" users
- "first requested" jobs

Other factors include: how you split the job, which machine you choose, etc.

Interpreting outcomes

Outcome visualization

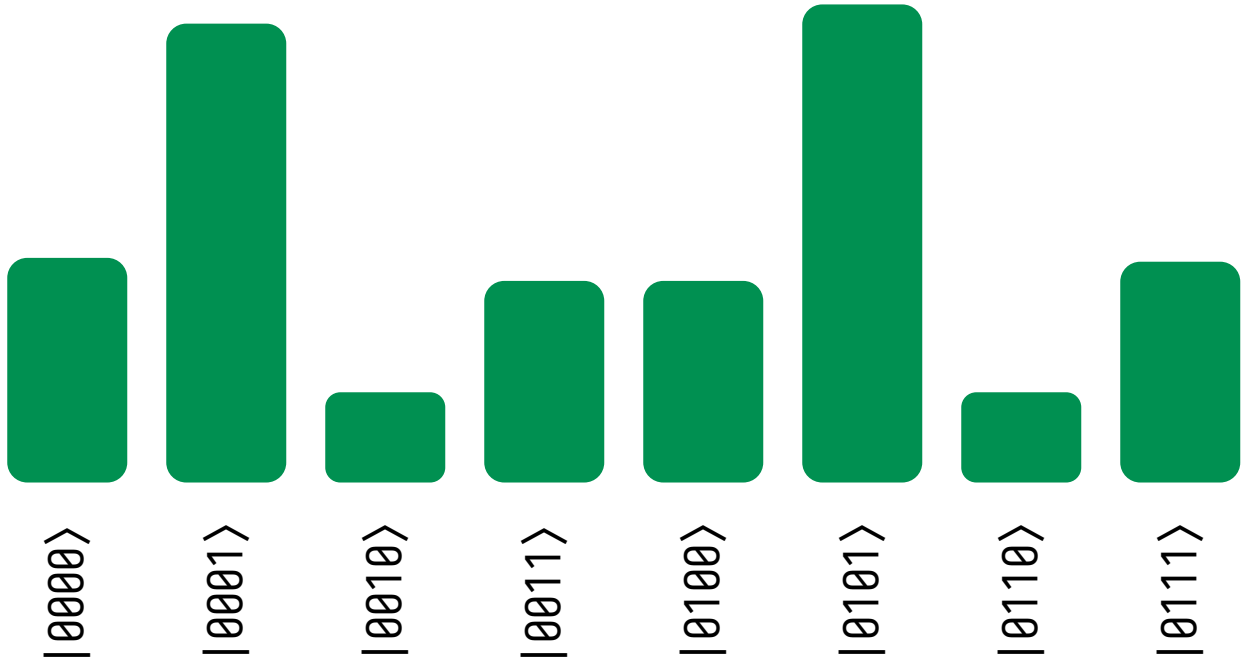
Noise analysis

Interpreting outcomes

Outcome visualization

Noise analysis

Typically, histograms



Problem-specific visualizations

[Kim et al., 2025]

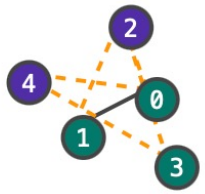
Logic solvers

State	a	b	c	d	Truth	Count
10100	F	F	T	F	T	372
10110	F	T	T	F	T	359
10010	F	T	F	F	T	212
10101	T	F	T	F	T	169
11010	F	T	F	T	T	143
10000	F	F	F	F	T	118
10001	T	F	F	F	T	111
10011	T	T	F	F	T	111
10111	T	T	T	F	T	95
11000	F	F	F	T	T	82
11001	T	F	F	T	T	62
11011	T	T	F	T	T	56
11110	F	T	T	T	T	56
11100	F	F	T	T	T	47

Graph solvers

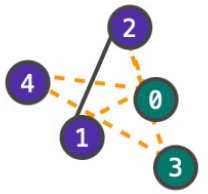
Candidate cut #1

- Bitstring (original): 10100
- Bitstring (reversed): 00101
- Count: 372
- # Cuts: 5
- Cuts: (0, 1, 3) / (2, 4)



Candidate cut #2

- Bitstring (original): 10110
- Bitstring (reversed): 01101
- Count: 359
- # Cuts: 5
- Cuts: (0, 3) / (1, 2, 4)



Interpreting outcomes

Outcome visualization

Noise analysis

Less of interest for end-users,
but more for NISQ era engineers

Mitiq | Error mitigation toolkit

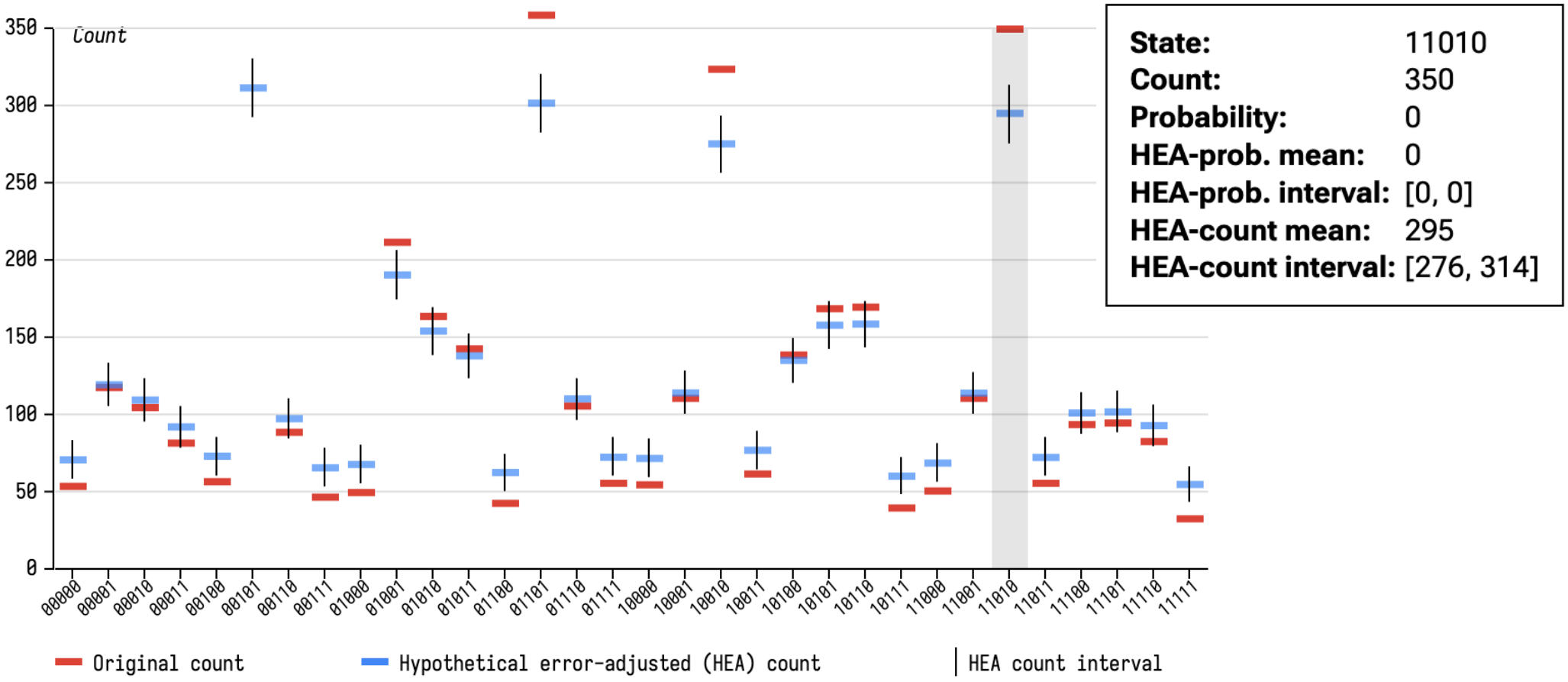
[LaRose et al., 2022]

Figure 4
https://quantum-journal.org/papers/q-2022-08-11-774/

With higher noise strength (p),
the energy level estimates appear stable
after error mitigation.

Hypothetical error adjustment

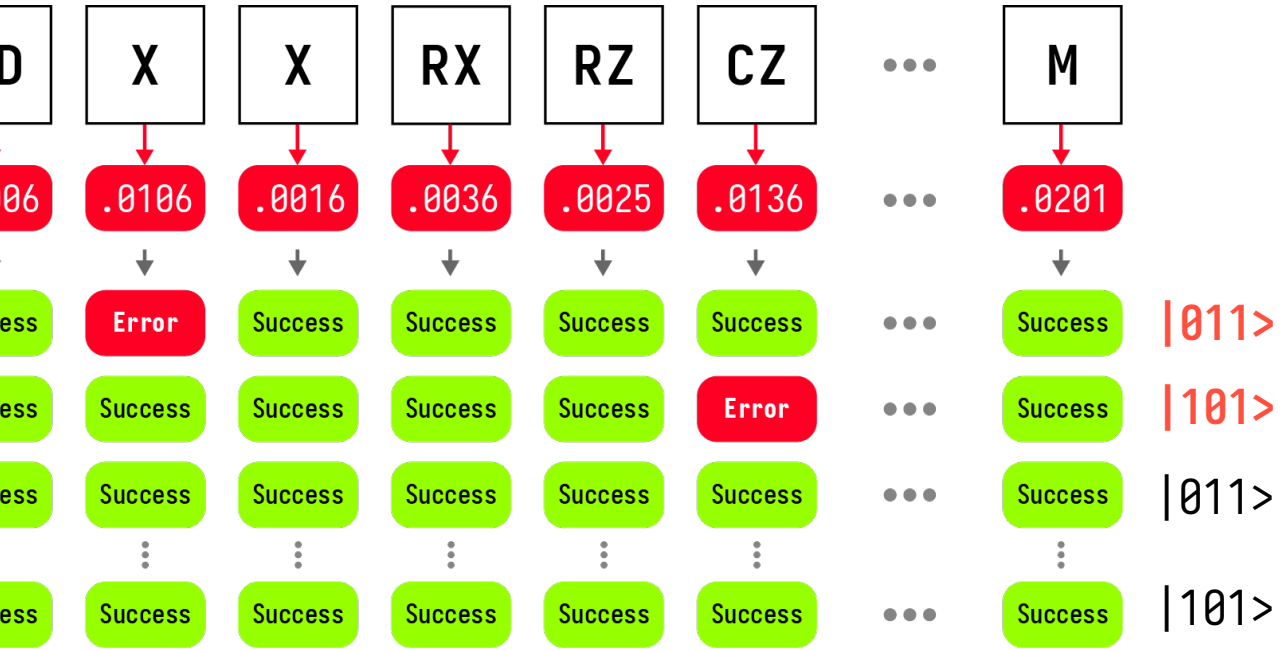
[Kim et al., 2025]



Providing uncertainty due to quantum noise

Simulating errors

[Kim et al., 2025]



Step 1 Retrieve the probabilities of operation errors

Based on calibrated error rates of the machine used for calculation

Step 2 Simulate the outcomes by applying the error probabilities

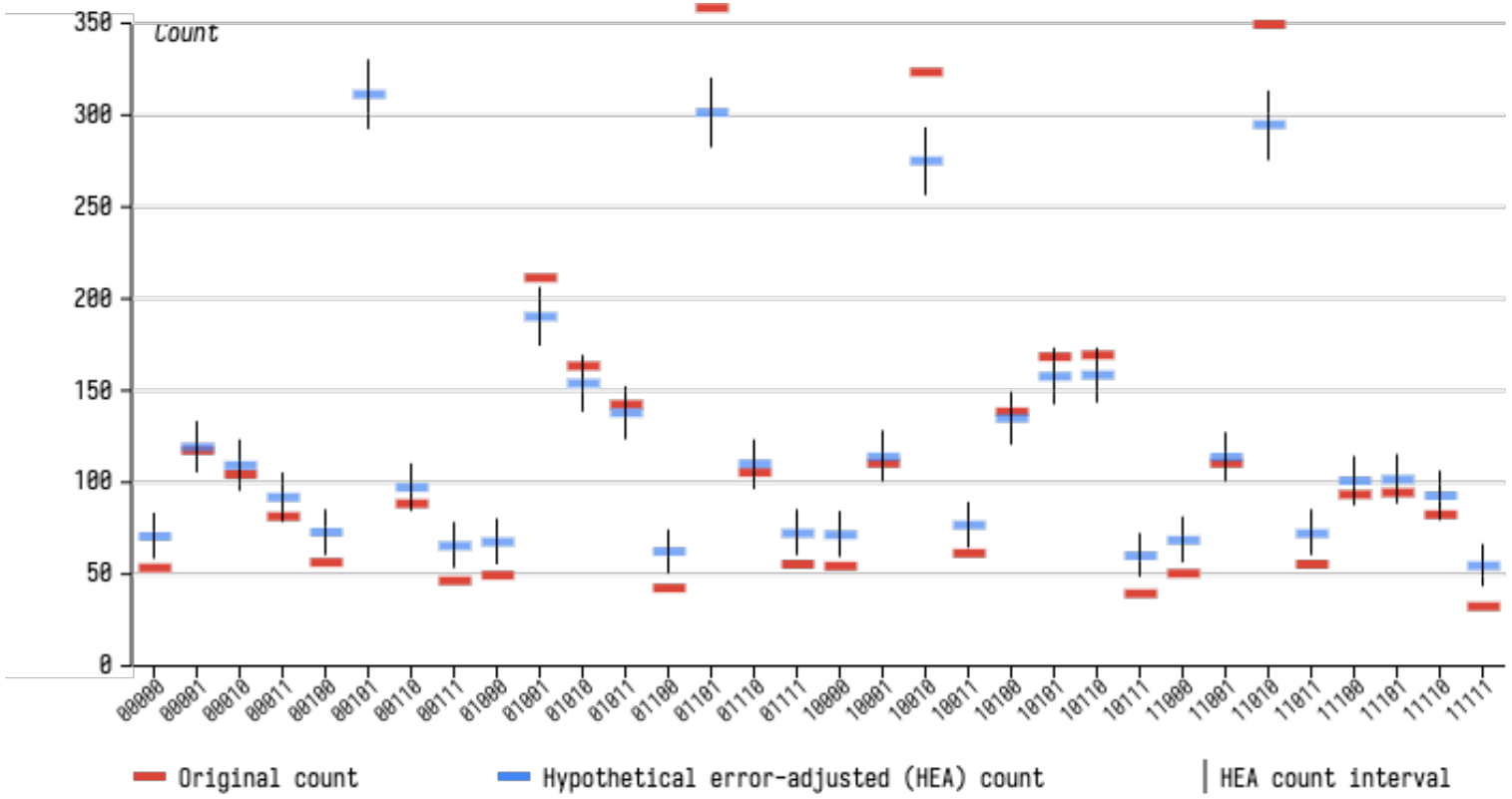
If an error is simulated, assign a different outcome based on the original distribution

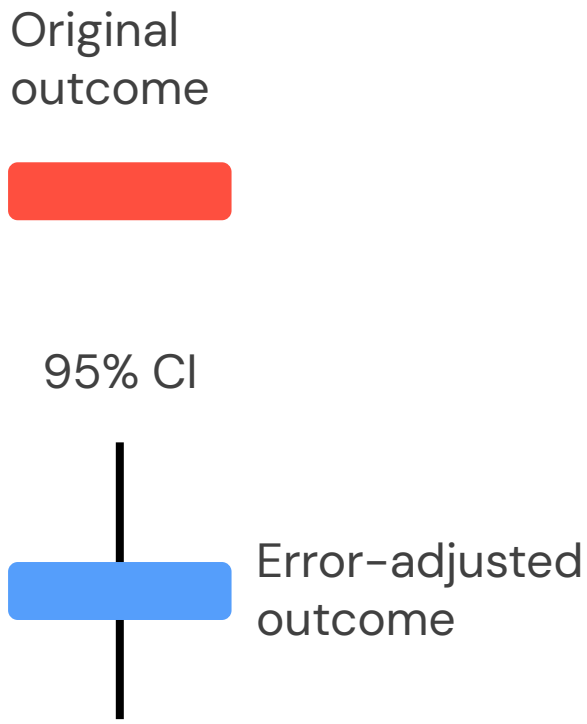
Step 3 Repeat a lot of times to obtain the distribution

Step 4 Plot the error distribution

Hypothetical error adjustment

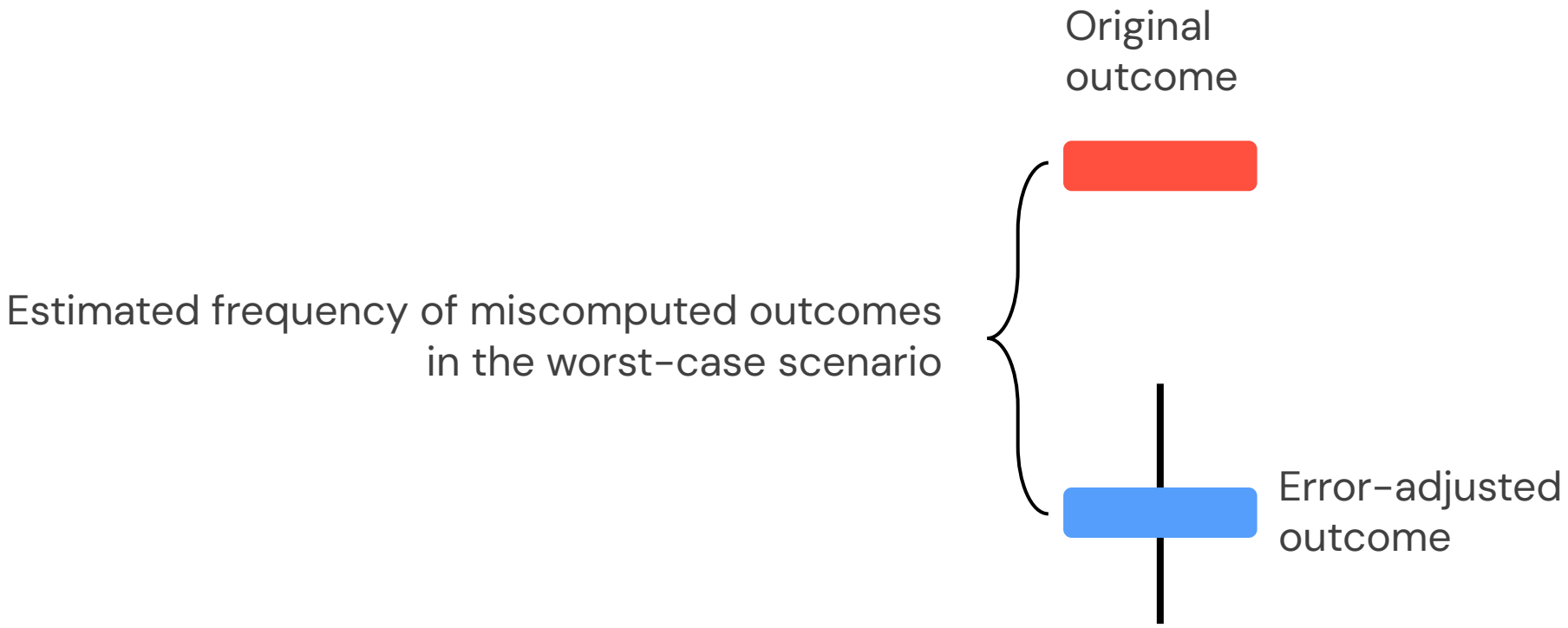
[Kim et al., 2025]





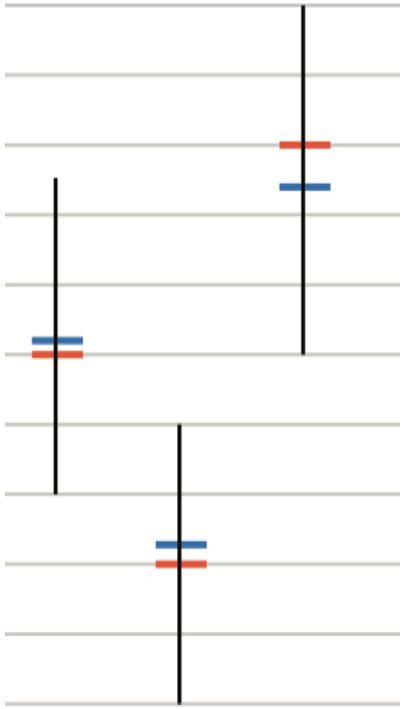
Hypothetical error adjustment

[Kim et al., 2025]



When we didn’t run the circuit enough times

[Kim et al., 2025]



CIs are overlapping
because the circuit was
not measured enough times.

Domain-specific approaches

Quantum machine learning

Quantum chemistry

VIOLET | Quantum Neural Network

[Ruan et al., 2024]

https://ieeexplore.ieee.org/abstract/document/10507082

Visualization for domain problems

[Microsoft]

```
from qdk_chemistry.data import Structure

structure = Structure.from_xyz_file('para-benzyne.structure.xyz')

structure
```

[1] ✓ 7.7s

```
Structure Summary:
  Number of atoms: 10
  Composition: H4, C6
  Total mass: 76.098 AMU
  Nuclear Repulsion Energy: 184.689 Eh
```

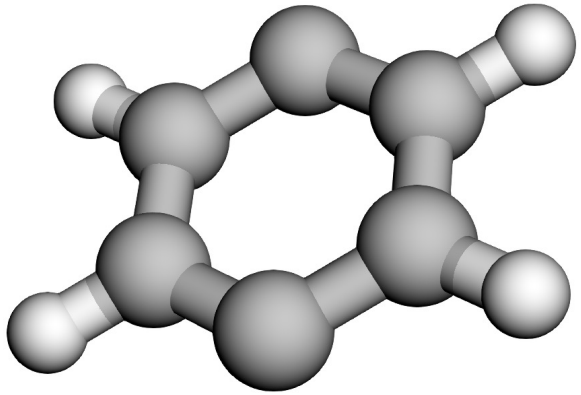
```
from qdk.widgets import MoleculeViewer

viewer = MoleculeViewer(molecule_data=structure.to_xyz())

viewer
```

[2] ✓ 0.0s

Visualization Style: Sphere



The molecular structure

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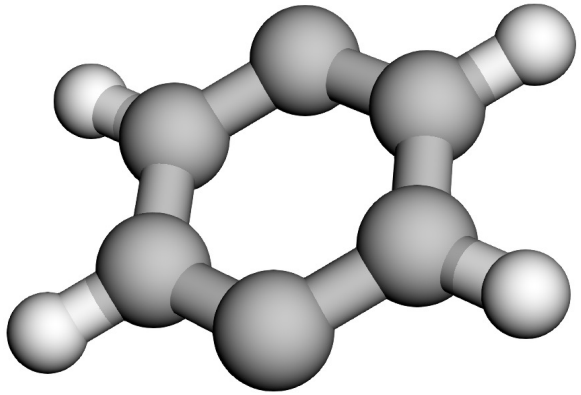
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```
File ~/Desktop/quantum/qdk-test/.venv/lib/python3
     73 dim = 1 << num_qubits  # 2^num_qubits
     75 # Initialize statevector as complex array
---> 76 statevector = np.zeros(dim, dtype=np.comp
     78 coeffs_array = np.array(coefficients)
     80 # Fill statevector

ValueError: Maximum allowed dimension exceeded
```

Can't process on my machine

What lacks here
How the molecular structure
is actually mapped to quantum circuits.

Challenge: not testable in real time

The script uses **22 qubits** as its primary working value — that's the minimal chemically meaningful active space: 11 spatial orbitals × 2 (spin-up + spin-down) = 22 spin-orbitals, mapped one-to-one to qubits under Jordan-Wigner.

The resource estimator in the code also computes the full-valence case: 22 spatial orbitals → **44 qubits**.

Here's the full table from the script's output:

System	Active space	Qubits
H_2 (for reference)	2 orb, 2 e	4
UO_2^{2+} minimal CAS	11 orb, 10 e	**22**
UO_2^{2+} full valence CAS	22 orb, 20 e	**44**

The 22-qubit version is technically simulable on a classical HPC cluster (statevector needs ~64 MB of RAM), but running 300 SPSA iterations against ~30,000 Pauli terms would still take hours. The 44-qubit version is infeasible classically — its statevector alone would need ~70 TB of RAM.

Prompt (summary)
Generate Qiskit code for VQE algorithm for uranyl dication

Education research for quantum computing

Highlights

Major focus is on early education: K-12 & undergraduate on basic concepts
Basic concepts: the half of what's covered today

Research tends to find that basic concepts are not that difficult to teach

Games could be useful to make the topic more familiar,
but students tend to find it difficult to map
the concepts from those games to those defined in quantum computing

Some considerations in motivating research problems in HCI/VIS for QC

Term

Near-term	Won't be useful with 100% FT QCs, but necessary for NISQ QCs
Mid-term	Applicability isn't clear with FT QCs, still useful for large-scale NISQ QCs
Long-term	Highly useful for application-scale FT QCs, but not achievable now

Users

End-users	Mainly interact with QCs via APIs, may not need device-level information (including researchers not in QC domain)
Software	Build software-wise infrastructure (error correction, etc.) for QC
Hardware	Build physical quantum computers heavily interact at a device level

FT: Fault-tolerant
NISQ: Noisy intermediate-scale QC

Some future paths

Rough examples

Quantum programming education

Materials, software, curriculum

Long End.

Quantum circuit visualization

Coordinating time and space

Near All

Easy-to-specify quantum programs

Could declarative programming help?

Mid End.

Domain-specific quantum software

Chemistry, physics, drug, finance, etc.

Long End.

Photonic device manipulation

High degree of freedom

Mid Soft Hard

Quantum error correction interface

Making QEC more flexible

Mid Soft

Decision support for QPU selection

Queue length, size, error profile, etc.

Near End.

Large-scale QPU/simulator visualization

Architecture research; High granularity

Near Mid Soft

Benchmark platforms for QC

QPU configs, routing methods, etc.

Near Soft Hard

Interpretability for AI-assisted quantum codes

Is the code verified theoretically?

Near End.